KUSAL DEY

Barrackpore, Kolkata-700122
kusaldey2004@gmail.com
7439689051

ABOUT ME

Information Technology graduate with a strong foundation in programming and software development. Proficient in C, Python, and Java, with a focus on Data Analytics, Machine Learning, Artificial Intelligence, and Generative AI. Applies analytical thinking, statistical modeling, and data-driven techniques to develop scalable, intelligent, and AI-powered solutions.

TECHNICAL SKILLS

Programming Languages : C, Java, Python
Other Languages : HTML, CSS, SQL
Python Libraries : Numpy, Pandas, Seaborn, Matplotlib
Frameworks and Databases : Tailwind CSS, ReactJS, NodeJS, MongoDB
Version Tools : Git
Other Developer Tools : VS-Code, Render, Vercel, Google Stitch, Antigravity
Core Subjects : Data Structures and Algorithms(DSA), Object Oriented Programming (OOPS), DBMS

KEY SKILLS

Problem-solving, Self-motivated, Team Management, Strong analytical skill, Quick Learner

EDUCATION

Bachelor of Technology (B.Tech) in Information Technology — July 2023 - Present
College : Narula Institute of Technology | CGPA : 8.94 (upto 5th semester)

Higher Secondary Examination — March 2021 - March 2022
School Name : Army Public School Barrackpore (CBSE) | Score: 80.80%

Secondary Examination — March 2019 - March 2020
School Name : Army Public School Barrackpore (CBSE) | Score: 91.60%

PROJECTS

VectorDB Engine — Custom AI Search & RAG Pipeline — Jan 2026 - Mar 2026
Python, FastAPI, NumPy, Docker, Ollama, HNSW, RAG
Architected a secure, containerized Vector Database from scratch using FastAPI and NumPy. Implemented custom HNSW indexing and Cosine Similarity, achieving sub-10ms semantic search latency on a foundational 200-record dataset to successfully validate retrieval efficiency. Engineered a hybrid search algorithm and integrated local LLMs via Ollama to power a fully offline, Docker-deployed Retrieval-Augmented Generation (RAG) pipeline.

EquiShare — Full-Stack Shared Expense Tracker — Sep 2025 - Nov 2025
Next.js, React, Convex, Inngest, Tailwind CSS, Shadcn/UI, Clerk, Gemini AI, Resend
Developed a full-stack financial dashboard using Next.js and Convex for real-time expense syncing. Implemented a custom debt-simplification algorithm—utilizing directed graph cash flow minimization—to net-out mutual balances, reducing total required group settlement transactions by up to 50%. Integrated Inngest cron jobs to automate state management and asynchronous tasks.

CERTIFICATIONS

SAP S/4HANA Development
NPTEL - Programming in JAVA
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

LANGUAGES

English (Fluent)
Bengali(Native)
Hindi (Fluent)